

Data Collection and Preprocessing Phase

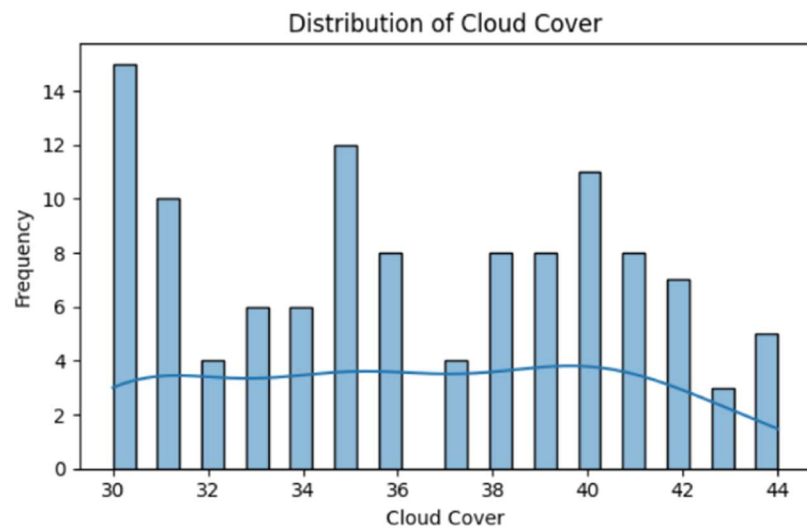
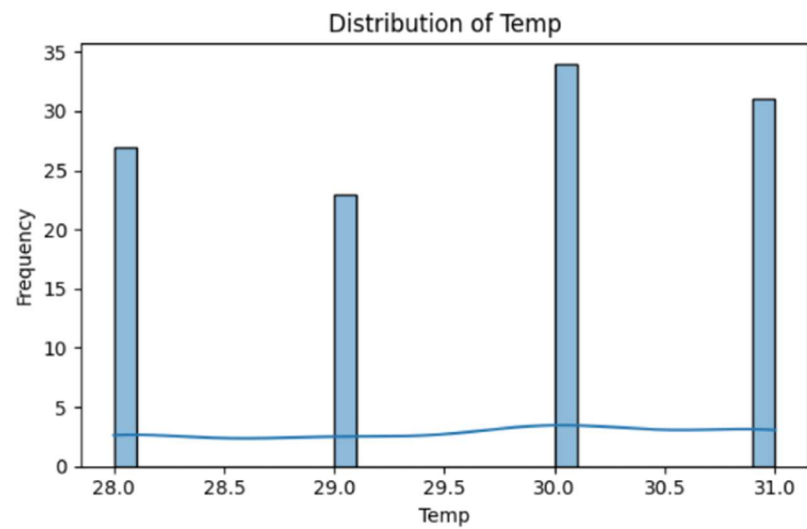
Date	02 Feb 2026
Team ID	LTVIP2026TMIDS75146
Project Title	Rising Waters: A Machine Learning Approach To Flood Prediction
Maximum Marks	6 Marks

Data Exploration and Preprocessing Template

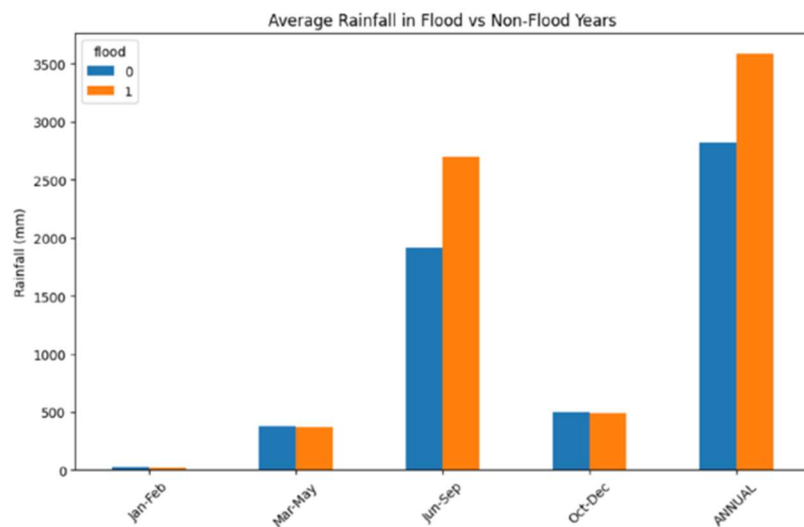
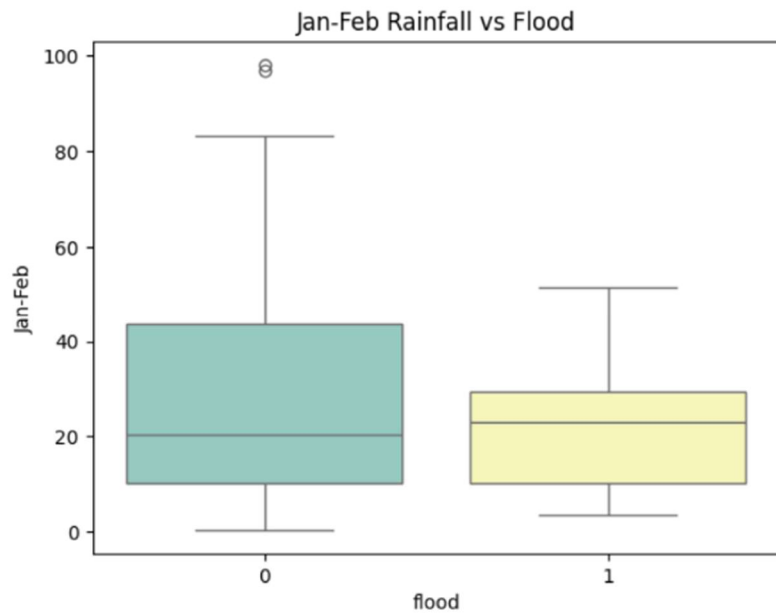
Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	Dimension: 115 rows x 11 columns Descriptive statistics:

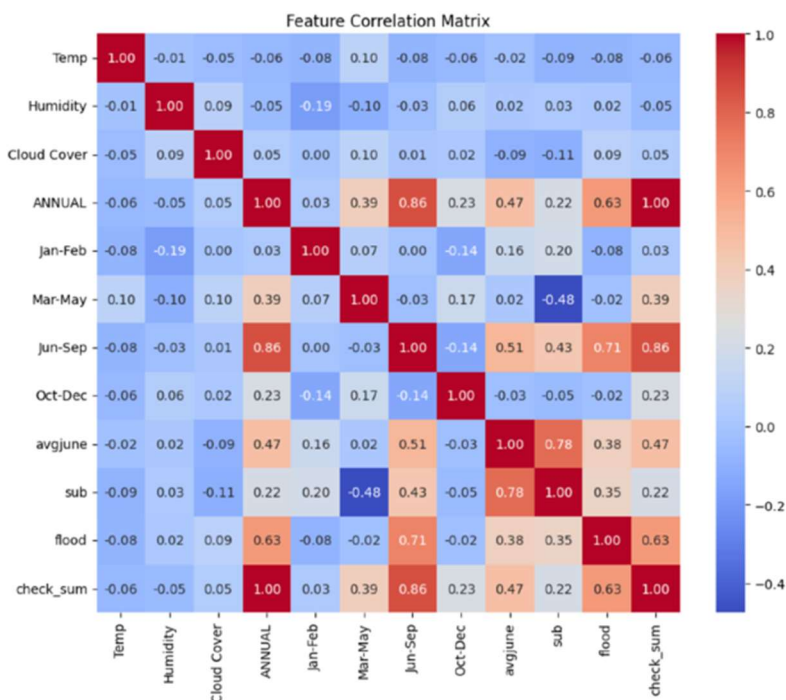
Univariate Analysis



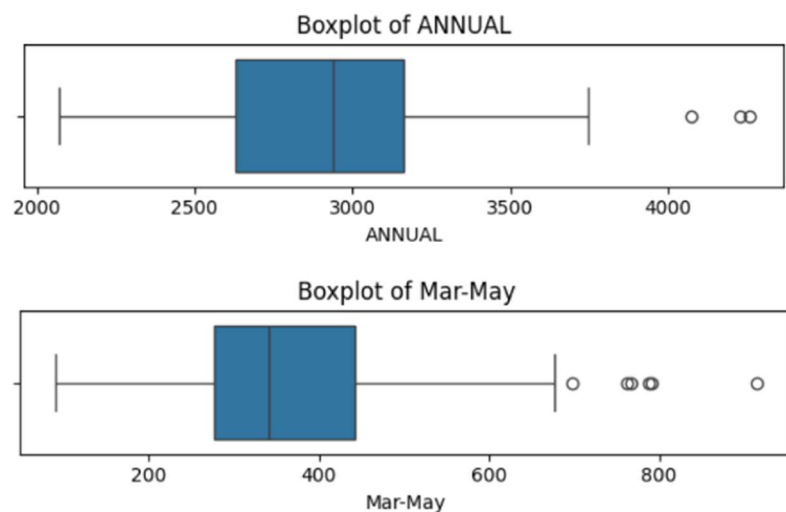
Bivariate Analysis



Multivariate Analysis



Outliers and Anomalies



Data Preprocessing Code Screenshots

Loading Data

```
df = pd.read_excel('../data/flood_dataset_raw.xlsx')
```

```
df.head()
```

	Temp	Humidity	Cloud Cover	ANNUAL	Jan-Feb	Mar-May	Jun-Sep	Oct-Dec	avgjune	sub	flood
0	29	70	30	3248.6	73.4	386.2	2122.8	666.1	274.866667	649.9	0
1	28	75	40	3326.6	9.3	275.7	2403.4	638.2	130.300000	256.4	1
2	28	75	42	3271.2	21.7	336.3	2343.0	570.1	186.200000	308.9	0
3	29	71	44	3129.7	26.7	339.4	2398.2	365.3	366.066667	862.5	0
4	31	74	40	2741.6	23.4	378.5	1881.5	458.1	283.400000	586.9	0

Handling Missing Data

```
df = pd.read_excel('../data/flood_dataset_raw.xlsx')
df.isnull().any()
```

```
Temp          False
Humidity       False
Cloud Cover   False
ANNUAL         False
Jan-Feb       False
Mar-May       False
Jun-Sep       False
Oct-Dec       False
avgjune       False
sub           False
flood         False
dtype: bool
```

Data Transformation

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Feature Engineering

Attached the codes in final submission.

Save Processed Data	-
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